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Piloted jet flames of CH_4/H_2/air: Experiments on localized extinction in the near field at high Reynolds numbers

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ABSTRACT

Measurements of temperature and major species concentrations, based on the simultaneous line-imaged Raman/Rayleigh/CO-LIF technique, are reported for piloted jet flames of CH_4/H_2 fuel with varying amounts of partial premixing with air (jet equivalence ratios of ϕ_j = 3.2, 2.5, 2.1 corresponding to stoichiometric mixture fraction values of ξ_{st} = 0.35, 0.43, 0.50, respectively) and varying degrees of localized extinction. Each jet flame is operated at a fixed and relatively high exit Reynolds number (60,000 or 67,000), and the probability of localized extinction is increased in several steps by progressively decreasing the flow rate of the pilot flame. Dimensions of the piloted burner, originally developed at Sydney University, are the same as for previous studies. The present measurements complement previous results from piloted CH_4/air jet flames as targets for combustion model calculations by extending to higher Reynolds number, including more steps in the progression of each flame from a fully burning state to a flame with high probability of local extinction, and adding the degree of partial premixing as an experimental parameter. Local extinction in these flames occurs close to the nozzle near a downstream location of four times the jet exit diameter. Consequently, these data provide the additional modeling challenge of accurately representing the initial development of the reacting jet and the near-field mixing processes.
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1. Introduction

Recent progress in fundamental understanding of turbulent combustion processes and in quantitative evaluation of computational combustion models has been accelerated by the availability of detailed measurements on well-defined flames [1,2] and by comparisons of measurements and modeled results based upon different modeling approaches. Comparisons have been conducted both within the collaborative framework of the International Workshop on Measurement and Computation of Turbulent Nonpremixed Flames (TNF) [1] and by individual researchers or groups. Among the various nonpremixed and partially premixed flames considered as modeling targets, a series of piloted jet flames of partially premixed CH_4/air [3–7] has received the greatest attention over the past decade and has been modeled using a range of methods and submodels (e.g., [8–16]). Parametric calculations of these piloted flames by Cao and Pope [14] have been particularly useful with respect to understanding of the sensitivity of computational results to changes in boundary conditions and submodels.

The piloted burner was originally developed at Sydney University [17,18] and variants on that design have been used to investigate localized extinction in several flames of various fuels and fuel combinations [2,19]. Localized extinction occurs when turbulent mixing rates between fuel and air become competitive with critical rates of chemical reaction. The annular pilot flame anchors the main jet flame at the jet exit, which has diameter d = 7.2 mm. Local extinction can occur after hot products of the pilot flame have been entrained into the jet, and cold coflowing air begins to be entrained and rapidly mixed with the jet fluid. The probability of local extinction increases as the main jet velocity increases (or the pilot flow decreases) until the upper portion of the flame ultimately blows off. Accurate prediction of the probability of local extinction and the associated trends in measured scalars is an important validation test for combustion models, and the Sandia piloted CH_4/air jet flames have been used extensively for this purpose.

As useful as these flames have been, they also have some limitations. One limitation is that the range of Reynolds number covered by the CH_4/air flame series begins at a relatively low value (Re = 13,400, 22,400, 33,600, 44,800 for flames C, D, E, F, respectively) [3]. Differential diffusion effects are apparent in some measurements, especially near the nozzle in the lower Re flames [7]. This causes some ambiguity in the interpretation of comparisons with computations because most models assume equal diffusivi-

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ties (unity Lewis number). Flames at higher Re would be desirable but cannot be achieved with partially premixed CH_4/air and coflowing air, both starting at room temperature. Another limitation is that flame F is very close to blowoff and is therefore highly sensitive to small changes in boundary conditions or model parameters. Flame D has low probability of local extinction, so only flame E has an intermediate level of extinction and additional intermediate steps would facilitate better model validation.

The present experiments were designed to address these limitations by moving to higher Reynolds number and including multiple steps in each progression from a fully connected flame to one with high probability of local extinction. Increased Re numbers were achieved by using a fuel blend of CH_4/H_2 (equal parts by volume) partially premixed with air. Augmentation of methane by hydrogen is also of potential interest in the context of alternative fuels and the burning of hydrogen rich fuel streams resulting from carbon sequestration technologies. Three different levels of partial premixing were used. Reduction of the equivalence ratio of the central jet increased the stoichiometric value of mixture fraction for the jet flame, and this shifts the location of the reaction zone inward relative to the turbulent shear layer in the near field of the jet. This serves as an additional test of the robustness of models. A controlled variation of partial premixing is also of practical interest because flames in combustors are often detached from surfaces, such that they burn as lifted flames with significant fuel–air mixing prior to combustion [20].

The premixed pilot flame burned CH_4/H_2 (equal parts by volume) and air at equivalence ratio 0.75, so inflow conditions may be represented using a single mixture fraction definition. Flow conditions were selected to produce fully connected turbulent jet flames with a relatively high pilot flow rate. Then the jet flow rate (and Reynolds number) was maintained constant, while the pilot flow was successively reduced, causing the probability of local extinction to increase. A total of 18 flames were investigated, six for each equivalence ratio in the main jet.

Lindstedt et al. [21] reported calculations of two of the flame series ($\phi_j = 3.2$ and $\phi_j = 2.1$) using a transported joint probability density function (PDF) approach closed at the composition-enthalpy level. The study focused on the impact of closure approximations for the scalar dissipation as applied to these near-field conditions, and it was shown that good agreement with measurements could be obtained by applying an extended algebraic closure for the scalar dissipation rate that accounts for small scale properties. The work also indicated that the current partially premixed flames may be treated in a manner that is consistent with fully premixed turbulent flames [22].

The present paper provides a full description of the experimental results for all of the 18 flames. Measurements within the pilot entrainment region ($x/d = 1$ and $x/d = 2$) are documented in addition to the main results on the extinction behavior at $x/d = 4$, where x is streamwise distance from the jet exit plane. The progression of increasing local extinction in each flame series is examined using scatter plots and density-weighted conditional means of selected scalars. The scalar structure of the fully burning flames is compared to calculations of laminar partially premixed flames over a range of strain rates with either multi-component molecular transport or the simplification of equal diffusivities (unity Lewis number). These comparisons indicate that preferential diffusion influences the scalar structure in the near field of these CH_4/H_2 flames, even at these relatively high Reynolds numbers. It is also apparent that high mixing rates in the fuel-rich portion of the flames preclude any significant progress of rich 'premixed' reactions that are important to the structure of strained laminar partially premixed flames.

2. Experimental methods

2.1. Diagnostics and flame conditions

Experiments were performed using the apparatus described in Ref. [5,6] for simultaneous line imaging of Raman scattering, Rayleigh scattering, and two-photon laser induced fluorescence (LIF) of CO. The combined system allowed measurement of single-shot profiles of temperature and the mass fractions of all major species (CO_2, O_2, CO, N_2, CH_4, H_2O, and H_2) with a spacing of 0.20 mm along a ~6 mm section of the focused laser beam. The estimated optical resolution was 0.3 mm. Total energy in the test section for the Raman/Rayleigh measurements was 1.6 J/pulse at 532 nm. Details of the hardware configuration are not repeated here.

Experiments were performed on piloted partially premixed CH_4/H_2/air flames having equal volume fractions of methane and hydrogen. Three mixture equivalence ratios were investigated, $\phi_j = 2.1$ (Re = 67,000), $\phi_j = 2.5$ (Re = 67,000), and $\phi_j = 3.2$ (Re = 60,000). Stoichiometric values of the mixture fraction for these flames are $\xi_{st} = 0.50$, 0.43, and 0.35, respectively. The piloted burner had the same geometry as used in previous experimental studies of partially premixed CH_4/air flames [3–6,23]. The central jet had exit diameter $d = 7.2$ mm. The inner and outer diameters of the pilot annulus were 7.7 mm and 18.2 mm, respectively, and the outer diameter of the complete burner was 18.9 mm. The pilot flame was supplied with premixed flow of CH_4/H_2 (1:1) and air at $\phi = 0.75$. The pilot velocity was varied to control the degree of localized extinction. Six different pilot flows were used with each of the three jet compositions for a total of 18 flames (Table 1). A laminar coflow of air at 0.9 m/s was supplied by a wind tunnel contraction with 5.8-to-1 area ratio and a 25 by 25 cm exit cross-section. The burner was mounted to extend roughly 15 cm above the exit plane of the wind tunnel.

With these flame conditions extinction/reignition occurred relatively close to the jet exit in a region centered near $x/d = 4$. In order to characterize the near-field region of each flame, radial profiles were obtained at $x/d = 1$ and 2, based on 100 shots at each of four overlapping positions of the 6-mm probe. Longer records of 2000 shots (nearly 60,000 individual samples) were obtained in the extinction region of each flame ($x/d = 4$) with the probe centered at $r = 6$ mm in the middle of the reacting shear layer.

2.2. Data processing, experimental limitations, and uncertainty

Processing of the Raman, Rayleigh, and CO-LIF line images was carried out as described in Ref. [6]. After background subtraction and normalization, the Rayleigh and CO-LIF results were mapped onto the positions of the Raman superpixels. The signals from each of these combined superpixel elements were then converted to species concentrations and temperature using an iterative matrix inversion approach. Diagonal elements of the matrix included a calibration factor and a temperature-dependent bandwidth function for each species. Off-diagonal elements accounted for temperature-dependent crosstalk among the Raman scattering spectra of the major species, laser-excited fluorescence interferences from various higher hydrocarbons formed in the flame, and broadband flame luminosity. Calibration curves for each matrix element were expressed as polynomials in temperature, and they were based on measurements in cold flows, electrically heated flows, and premixed flat flames. Temperatures in the flat flames were measured separately using CARS, and equilibrium species concentrations were calculated at the measured temperatures (nonadiabatic) and laboratory pressure. The signal from the strong C_2 band near 560 nm was used to monitor hydrocarbon fluorescence, while signal near 695 nm was used as a measure of broadband luminosity,

Table 1
Flame designations and flow parameters.

Flames	ϕ_j	Jet flow[a] (lpm)	u_j (m/s)	Re ($u_j d/\nu$)	Pilot flow[b] and % of jet[c]					
					a	b	c	d	e	f
pmh32a-f	3.17	349	143	60,000	50	37	25	22	20	18
					4.6	3.4	2.3	2.0	1.8	1.6
pmh25a-f	2.50	384	157	67,000	50	37	25	22	20	18
					4.9	3.6	2.5	2.2	2.0	1.8
pmh21a-f	2.11	381	156	67,000	50	37	25	20	18	16
					5.6	4.2	2.8	2.2	2.0	1.8

[a] Total volumetric flow rate of CH_4/H_2/air at 1 atm, 294 K through the central jet.
[b] Total volumetric flow rate of CH_4/H_2/air at 1 atm, 294 K through the pilot at $\phi = 0.75$.
[c] Pilot enthalpy as a percentage of total enthalpy of the central jet is lists below the pilot flow rate.

as described below. Within each iteration the major species mole fractions from the Raman/LIF measurements were used to determine the species-weighted Rayleigh scattering cross-section, and the resulting Rayleigh temperature was applied in recalculating the temperature-dependent matrix elements until convergence to <1 K was achieved (typically 3–4 iterations).

This data set has some limitations that should be considered. First, because of the large number of flames and some operational constraints, measurements were limited to the near field. Second, based on what we know of turbulence scales in jet flames [24–27] the spatial resolution of the experiments is not sufficient to measure scalar dissipation in the present flames. Results for 'resolved' dissipation are reported as a qualitative indication of where the high-dissipation regions occur and what the trends in dissipation are. However, no effort has been made to estimate what fraction of true dissipation has been resolved, and this resolved fraction will vary significantly with radial location because the mean mixture fraction gradient varies significantly with radius near the base of the flame. Third, results indicate that scalar profiles are somewhat influenced by differential diffusion even though the Reynolds number is relatively high. Finally, these flames produce higher levels of chemiluminescence than the CH_4/air piloted flames, presenting some problems as explained below.

The current flames are characterized by lower levels of fluorescence interference but higher levels of flame luminosity than the previously studied CH_4/air flames. Normally, a CH_4/H_2 flame would be less luminous than a corresponding CH_4 flame. In the present work, however, the high Reynolds number yielded high rates of mixing and heat release, so chemiluminescence rates were correspondingly high. Crosstalk terms for fluorescence interference and luminosity cannot be calibrated directly but must be optimized based upon correlations within the turbulent flame data and physical constraints (i.e., avoiding negative conditional mean mass fractions resulting from over correction of these effects). Single-shot corrections for the contribution of luminosity to the measured Raman signals were based on the luminous background signal from a Raman-free portion of the spectrum near 695 nm to the red side of the H_2 vibrational Raman scattering band. However, these corrections did not work as well as hoped, because the spectrum of luminosity can vary from shot to shot and the signal is nonlocal (photons are collected with varying efficiency from all regions of the flame intersecting the view of the collection optics). The temporal gate for these Raman scattering measurements was triangular with full width of 9.2 μs at half maximum. This was achieved using the rotating mechanical shutter described in Ref. [28]. Faster gating of 3.9 μs has since been achieved using a new shutter design. However, Raman scattering measurements in highly luminous flames may require intensifier gating (at the expense of collection efficiency), a spectral fitting approach to data processing [29], or implementation of polarization separation and subtraction [30,31].

Corrections for luminosity were tuned as a best effort, based on the experience of the first author. The $\phi_j = 3.2$ and $\phi_j = 2.5$ flames were processed with the same 'calibration' file. The luminosity corrections were adjusted slightly for the $\phi_j = 2.1$ flames to eliminate negative conditional mean H_2 mass fraction values in the fully burnt case. Uncertainty in the corrections for flame luminosity contributes to the uncertainty in the measured mass fractions, but the precise contributions cannot be determined from the available data. These corrections may be too large or too small and can therefore cause errors in either direction in mixture fraction and individual species mass fractions. Because of the viewing angle of the collection lens, measurements in stoichiometric and rich samples are more influenced by luminosity than lean samples. Overall, we believe that uncertainties in species mass fractions and the mixture fraction are perhaps ~2% higher in these flames than in previous CH_4/air flames. Luminosity does not significantly increase uncertainty in temperature from Rayleigh scattering. This is because fluctuations in luminosity tend to affects various species similarly, such that mole fractions and the calculated Rayleigh scattering cross-section are not strongly influenced. Table 2 lists representative uncertainties for measured scalars. These uncertainties are still small compared to mass fraction variances in the locally extinguished flames and compared to the measured changes in temperature and species as local extinction probability increases. Therefore, the main value of these flames in testing predictions of localized extinction remains intact.

The mixture fraction was calculated from the measured mass fractions using the Bilger formulation [32] modified as in the previous work on partially premixed flames [3,5–7] to exclude the oxygen terms. This reduces the sensitivity of mixture fraction results to measurement noise and fluorescence interferences. Average values of mixture fraction obtained from the modified expression and from the full Bilger expression are nearly identical in the turbulent flames investigated here.

Table 2
Representative uncertainties in scalar measurements in premixed flat flames with low luminosity and in the turbulent jet flames with higher luminosity. Listings for the turbulent flames are increased from those for the flat flames to allow approximately for uncertainty in corrections for interference from flame luminosity. Values are representative of fuel-rich conditions at elevated temperature.

Scalar	Precision, σ (%)	Accuracy (flat flames, %)	Premixed flame	Accuracy (turbulent flames, %)
T	0.6	2	$\phi = 0.97$, $T = 2185$	2
N_2	0.7	2	$\phi = 0.97$, $T = 2185$	3
CO_2	3.0	4	$\phi = 0.97$, $T = 2185$	6
H_2O	2.2	3	$\phi = 0.97$, $T = 2185$	6
ξ	2.1	5	$\phi = 0.97$, $T = 2185$	8
CO	5	10	$\phi = 1.28$, $T = 2045$	10
H_2	7.5	10	$\phi = 1.28$, $T = 2045$	10

3. Results and discussion

For each series of flames the flow rate through the central jet remains constant while the flow rate to the pilot flame is decreased. In order to understand the changes in the overall flow field that lead to increased probability of localized extinction, scalar measurements from the pilot entrainment region are presented first and followed by a discussion of results from the extinction region.

3.1. Pilot entrainment region

Measurements of scalar profiles at the exit plane of the jet would be useful in defining boundary conditions for calculations, but cannot be obtained with the same completeness and accuracy as measurements slightly downstream. Therefore, in order to provide a check on modeled inflow conditions and track the pilot entrainment process upstream of the extinction region, radial profiles were measured at $x/d = 1$ and $x/d = 2$ in each flame. Each profile included a total of 400 laser shots, with 100 shots taken with the 6-mm imaged region centered at each of four locations, $r = 0$, 3, 6, 9 mm, overlapping by half. Radial profiles of Favre average mixture fraction and temperature are shown in Figs. 1–3 along with corresponding density weighed conditional means for the three different flame series.

It is immediately obvious in each flame series that unburnt reactants from the pilot flow are present at $r \sim 8$ mm near the outer edge of the pilot annulus in the cases having the highest pilot flow rates (50 and 37 lpm). This was not apparent during the experiments and was discovered only after the measurements were completed and data fully processed. That is to say the pilot flame looked normal to the eye for all cases. We do not consider this to be a significant issue for modeling of these flames for the following reasons. The fraction of pilot gas affected is small, and the unburnt fuel appears only near the outer edge of the pilot. Most of the fuel is burnt by $x/d = 2$, with only a small deviation in mean profiles for the 50-lpm cases remaining at that position. This burnout occurs well away from the region where the central jet is entraining burnt gas from the pilot. In all the flames with significant extinction, i.e., with pilot flows of 25 lpm and below, there is no indication of unburnt pilot gas at any measurement location. This was confirmed in the scatter data (not shown) where a small percentage of CH_4 mass fraction values reached 0.005 in the 37 lpm pilot cases at $x/d = 2$ but dropped to zero (within measurement uncertainty) for the 25 lpm and lower pilot cases. Even with the 50-lpm pilot flames, the progression of results from $x/d = 1$, 2, and 4 suggests that any pockets of pilot reactants burn out before being entrained into the central jet. We therefore propose that computations may neglect this detail and assume a fully burnt pilot flow at the exit plane.

Aside from the deviations caused by unburnt fuel near the outer part of the pilot flame, the density-weighted conditional mean profiles of temperature and CH_4 mass fraction, Y_{CH_4}, are nearly identical within each flame series, especially at $x/d = 1$. This is because the central jet velocity is fixed in each series, and nothing significant happens until the pure products of the pilot are fully entrained and the jet begins to entrain gas that has been diluted by coflowing air. This happens faster as the pilot flow rate is reduced. At $x/d = 2$ the plateau in the radial profiles of mixture fraction that corresponds to the pilot condition has disappeared in all but the highest pilot flow cases.

Consider the $\phi_j = 2.1$ flame series (Fig. 3) as an example. The maximum Favre average temperature decreases with decreasing pilot flow, and the steep portion of the mixture fraction profiles (centered near $r = 4.5$ mm) extends progressively to lower values of ξ before becoming more gradual. The corresponding conditional mean curves for T at $x/d = 2$ (also in Fig. 3) shift downward due to finite rate chemistry. Localized extinction is observed for a small fraction of samples in the scatter data (not shown) from the lowest

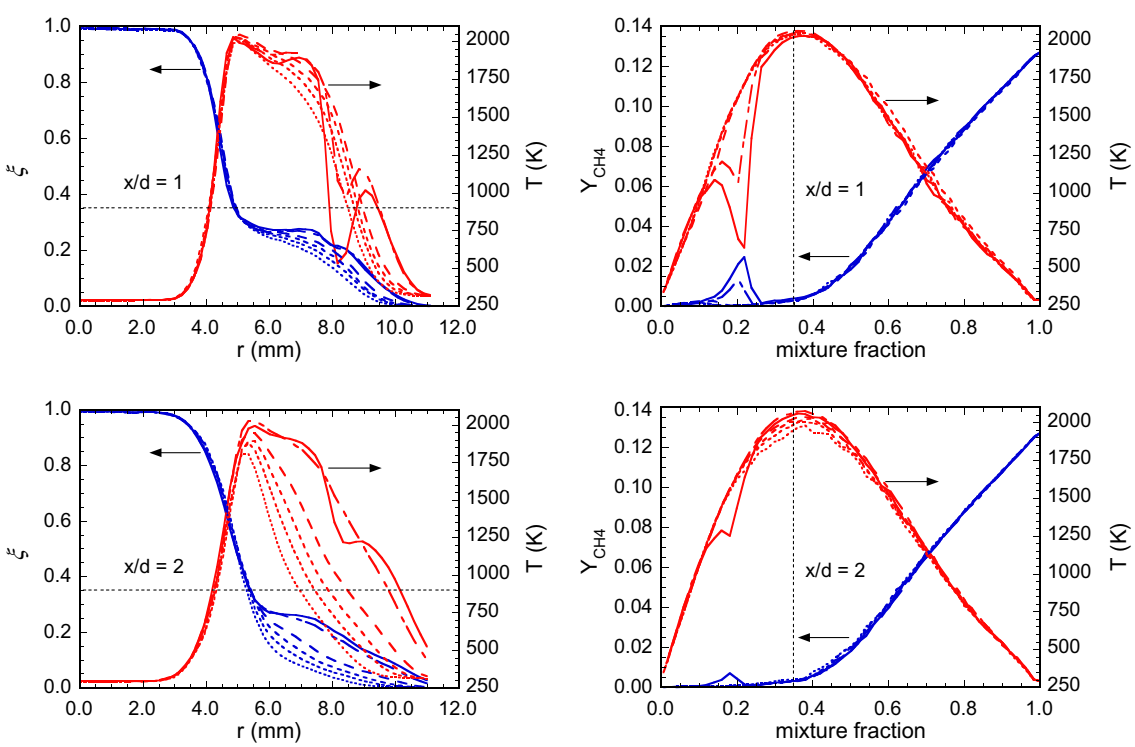

Fig. 1. Favre average radial profiles (left column) of mixture fraction and temperature and density-weighted conditional means (right column) of CH_4 mass fraction and temperature at $x/d = 1$ and $x/d = 2$ in the $\phi_j = 3.2$ turbulent flame series. Line types correspond to the six pilot flow rates listed in Table 1: 50 lpm (solid) down to 18 lpm (dotted). The stoichiometric mixture fraction $\xi_{st} = 0.35$ is marked by a dashed line in each graph.

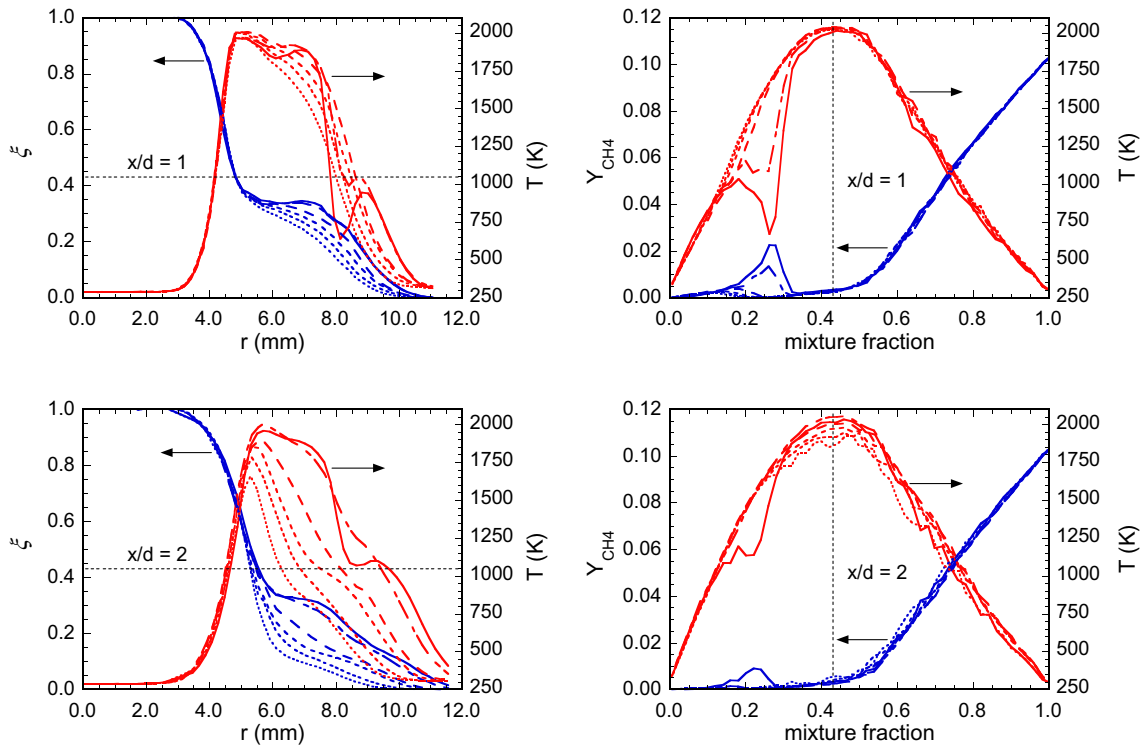

Fig. 2. Favre average radial profiles (left column) of mixture fraction and temperature and density-weighted conditional means (right column) of CH_4 mass fraction and temperature at $x/d = 1$ and $x/d = 2$ in the $\phi_j = 2.5$ turbulent flame series. Line types as in Fig. 1. The stoichiometric mixture fraction $\xi_{st} = 0.43$ is marked by a dashed line in each graph.

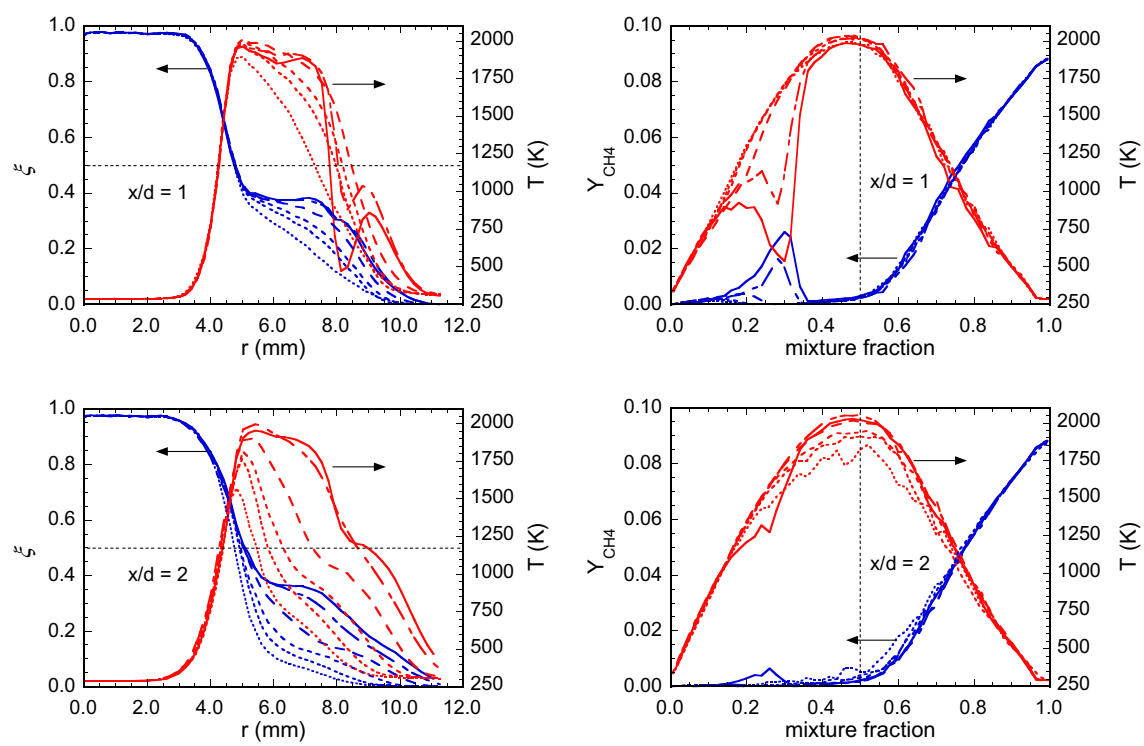

Fig. 3. Favre average radial profiles (left column) of mixture fraction and temperature and density-weighted conditional means (right column) of CH_4 mass fraction and temperature at $x/d = 1$ and $x/d = 2$ in the $\phi_j = 2.1$ turbulent flame series. Line types correspond to the six pilot flow rates listed in Table 1: 50 lpm (solid) down to 16 lpm (dotted). The stoichiometric mixture fraction $\xi_{st} = 0.50$ is marked by a dashed line in each graph.

pilot flow case and conditional mean results for Y_{CH_4} show a corresponding increase.

Fig. 4 includes Favre average radial profiles and density-weighted conditional means of the resolved one-dimensional sca-

lar dissipation rate, $\chi_r = 2D_\xi(\partial\xi/\partial r)^2$. Here, the mixture diffusivity was calculated from measured mole fractions and temperature as

$$D_\xi = 1.05 * \sum D_i X_i, \quad (1)$$

where

$$D_i = [(D_i \mid \xi = 0) + (D_i \mid \xi = 1)] * (T/298)^{1.67}.$$

The factor 1.05 was applied to improve agreement with diffusivity values calculated for a laminar flame using an Imperial College code [33] and the mechanism described in Ref. [21]. It is important to emphasize that the dissipation results reported here are only qualitative. Based on recent work on turbulence length scales and resolution requirements for dissipation measurements in jet flames [24–27], we can be sure that the present measurements are not fully resolved. We can also be reasonably assured that attenuation by spatial averaging within the high-dissipation regions of these flames is more significant than augmentation by experimental noise. Therefore, results shown here are expected to be lower than the true dissipation values in the high-dissipation regions. We refer to these as 'resolved' dissipation results, but have not attempted to estimate the fraction of the true dissipation rate that has been resolved. This fraction will vary significantly across the flames and locations considered.

The resolved dissipation profiles and conditional means in Fig. 4 may be related directly to the corresponding results in Fig. 3. At $x/d = 1$ there is almost no effect of pilot flow rate on the high-dissipation region. The maximum conditional mean dissipation rate occurs near $\xi = 0.6$ for all six pilot flows. With combustion products from the pilot flame located at the outer edge of the jet shear layer, the scalar dissipation rate in the reaction zone (near stoichiometric) remains low enough to allow combustion to proceed. At $x/d = 2$ the effect of pilot entrainment is much more obvious. As the pilot flow decreases the mixture fraction value and temperature at the outer edge of the high shear layer also decrease. When the pilot flow drops below 25 lpm the peak in the Favre average profile of scalar dissipation increases, and the conditional mean dissipation curve extends toward stoichiometric and lean mixture fraction values. For the lowest pilot flow the resolved conditional mean dissipation curve is centered near the stoichiometric value of mixture fraction, and this promotes the observed onset of localized extinction.

3.2. Region of local extinction

The measurement location at $x/d = 4$ was roughly at the middle of the visually observed region of reduced flame luminosity due to local extinction. None of the measured flames exhibited global extinction; all developed into fully burning flames farther downstream. At this near-field location the 6-mm probe segment was sufficient to include almost the entire mixing layer from potential core (pure jet fluid) to coflowing air. Consequently, all measurements (2000 shots for each flame condition) were acquired at just one position of the probe, which was oriented along the flame radius and centered 6 mm from the jet axis.

The progression of local extinction in the flame series with $\phi_j = 3.2$ ($\xi_{st} = 0.35$) is examined in Fig. 5 using scatter plots of Y_{CH_4} and T. Fig. 6 includes the corresponding radial profiles of Favre average ξ, T, and χ_r, as well as density-weighted conditional means of Y_{CH_4}, T, and χ_r. The results for $\phi_j = 2.5$ ($\xi_{st} = 0.43$) are shown in Figs. 7 and 8, and those for $\phi_j = 2.1$ ($\xi_{st} = 0.50$) are shown in Figs. 9 and 10.

In each flame series, the 50 lpm pilot case has only very few samples that deviate from the narrow band of temperature or CH_4 scatter points. Therefore, these flames may be considered fully connected, although they are still relatively highly strained, particularly in the fuel-rich part of the mixing layer, as will be shown later. Probability of local extinction increases gradually in each series of scatter plots as the pilot flow decreases.

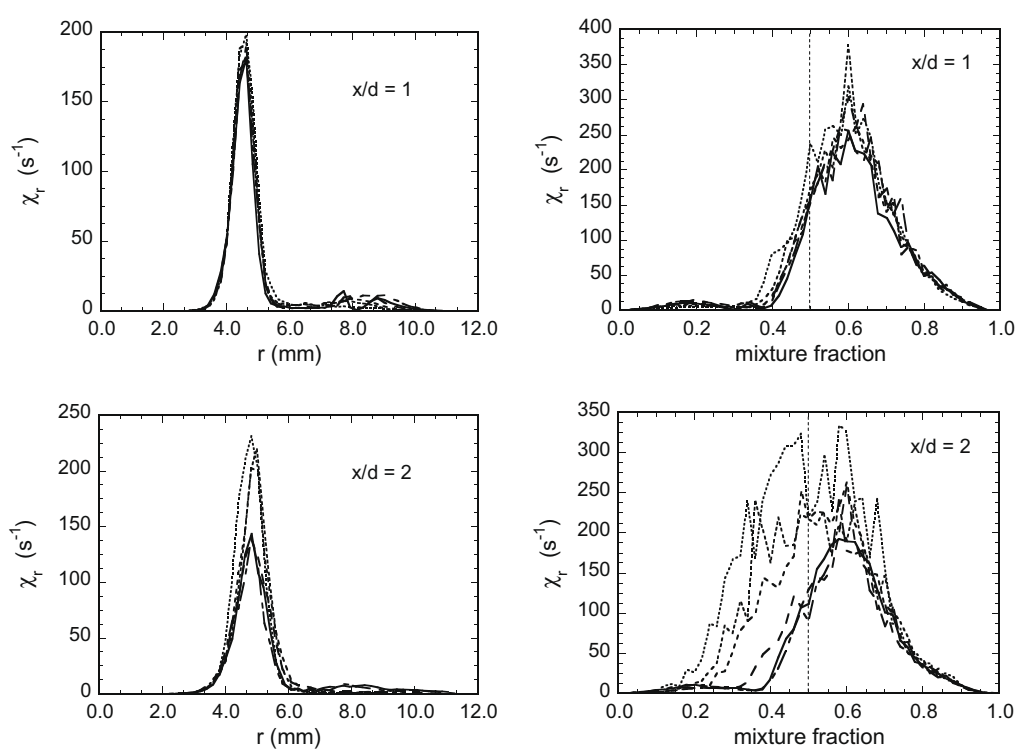

Fig. 4. Favre average radial profiles (left column) and density-weighted conditional means (right column) of the resolved radial scalar dissipation at $x/d = 1$ and $x/d = 2$ in the $\phi_j = 2.1$ turbulent flame series. Line types as in Fig. 3.

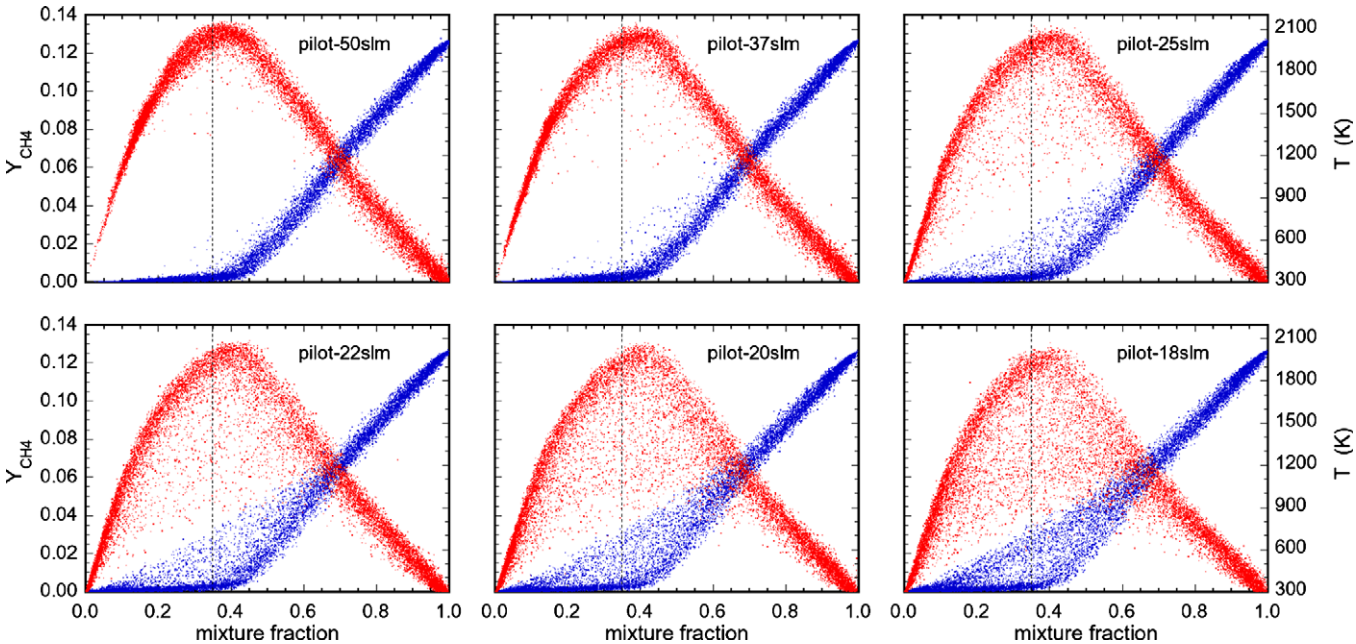

Fig. 5. Scatter plots of CH_4 mass fraction and temperature at $x/d = 4$ in each of the six flame with $\phi_j = 3.2$. The stoichiometric mixture fraction $\xi_{st} = 0.35$ is marked by a dashed line in each graph.

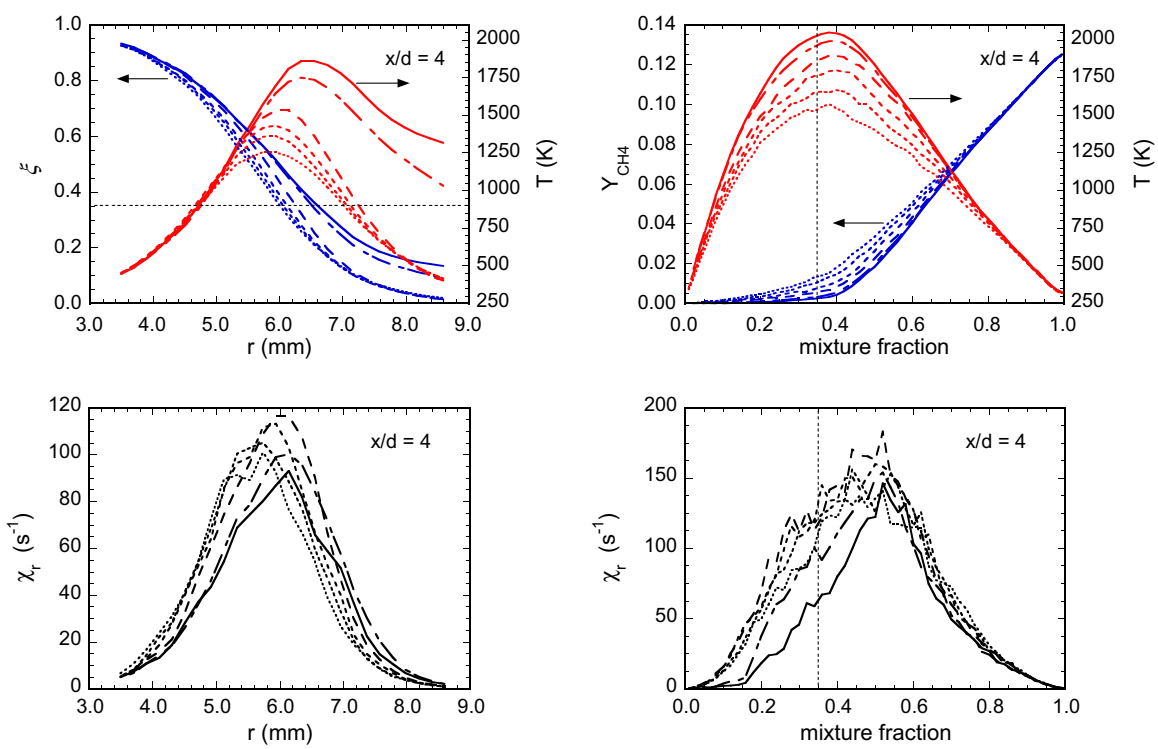

Fig. 6. Favre average radial profiles (left column) and density-weighted conditional means (right column) including mixture fraction, temperature, CH_4 mass fraction, and resolved scalar dissipation at $x/d = 4$ in the $\phi_j = 3.2$ flame series. Line types correspond to the six pilot flow rates listed in Table 1: 50 lpm (solid) down to 18 lpm (dotted). The stoichiometric mixture fraction $\xi_{st} = 0.35$ is marked by dashed lines.

Radial profiles of ξ and T reveal a dramatic narrowing of the overall mixing layer as a result of the first two changes in pilot flow rate. The increase in the probability of local extinction is modest with each of these two large steps in pilot flow (from 50 to 37 lpm and then to 25 lpm). The probability of extinction then becomes much more sensitive to further reductions in pilot flow. While the measured dissipation is not fully resolved, qualitative trends still yield useful insights. Results for the conditional mean scalar dissipation rate suggest that the observed sensitivity to pilot flow rate comes when dissipation within the reaction zone reaches a critical level. In each flame series there are significant increases in the stoichiometric value of the resolved scalar dissipation accompanying the first two downward steps in pilot flow rate. Subsequent, smaller reductions in pilot flow, cause significant increase

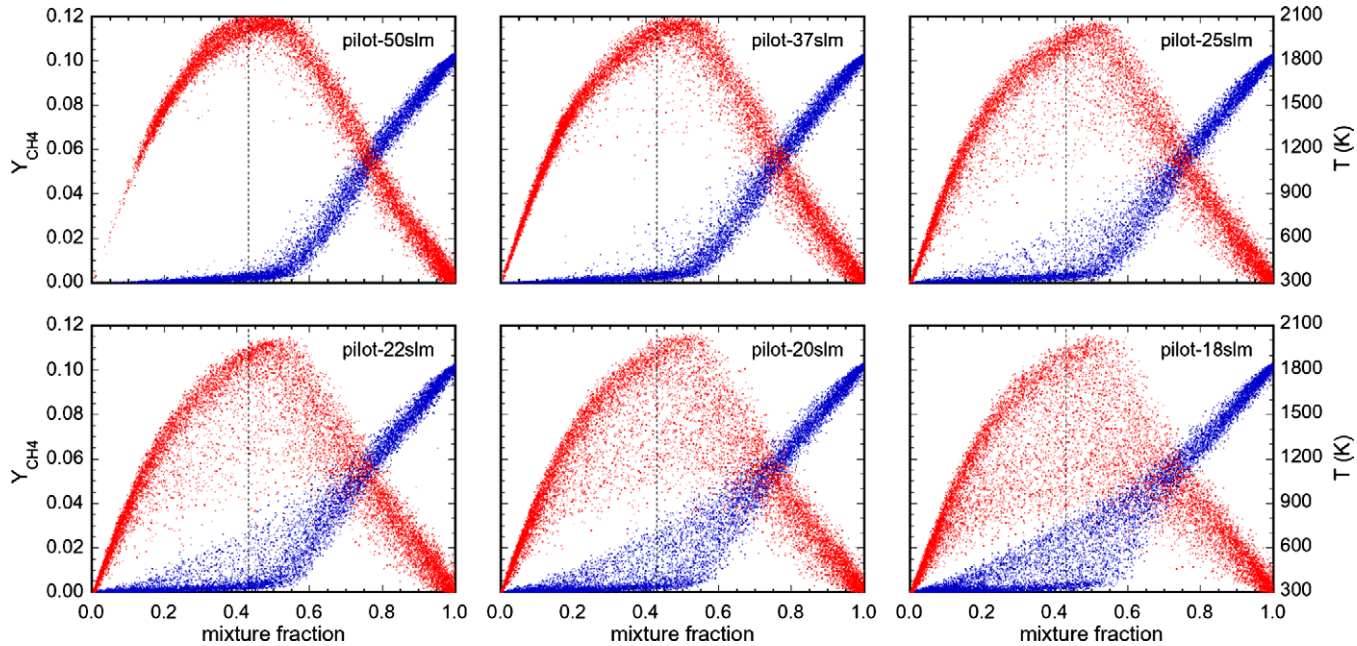

Fig. 7. Scatter plots of CH_4 mass fraction and temperature at $x/d = 4$ in each of the six flame with $\phi_j = 2.5$. The stoichiometric mixture fraction $\xi_{st} = 0.43$ is marked by a dashed line in each graph.

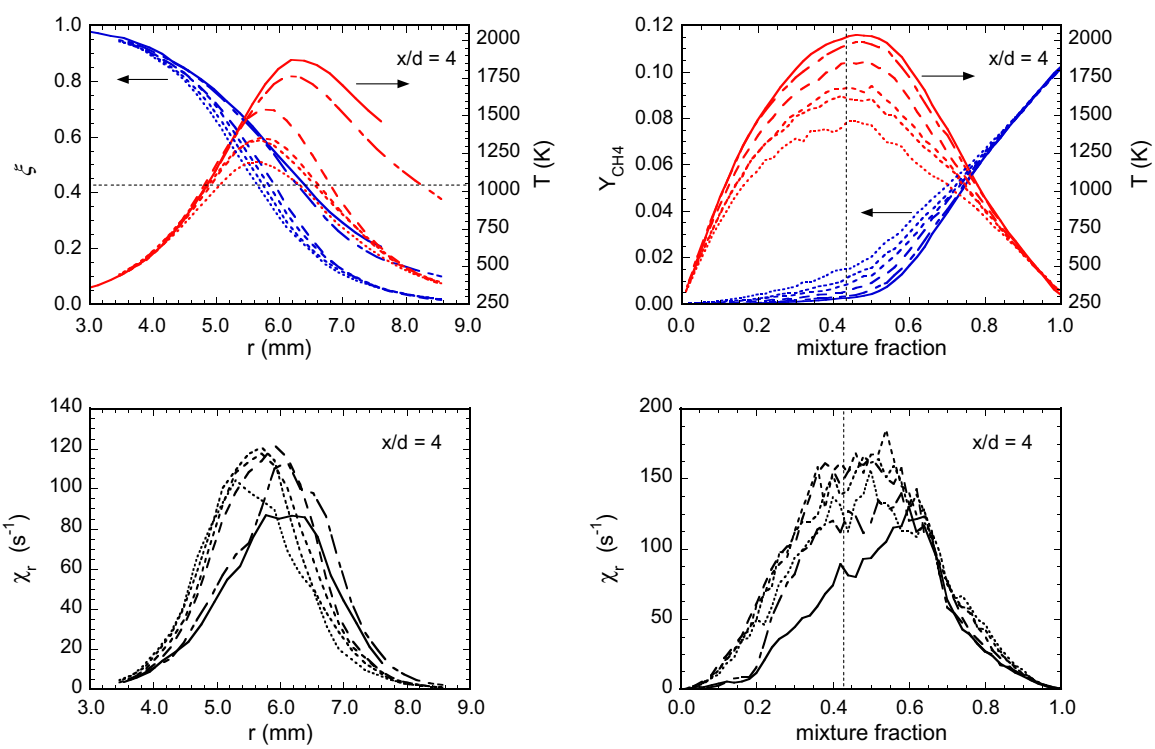

Fig. 8. Favre average radial profiles (left column) and density-weighted conditional means (right column) including mixture fraction, temperature, CH_4 mass fraction and resolved scalar dissipation at $x/d = 4$ in the $\phi_j = 2.5$ flame series. Line types correspond to the six pilot flow rates listed in Table 1: 50 lpm (solid) down to 18 lpm (dotted). The stoichiometric mixture fraction $\xi_{st} = 0.43$ is marked by dashed lines.

in the probability of extinction but only minor changes in conditional mean scalar dissipation. In each flame series the conditional mean dissipation curve clearly moves downward for the final, lowest pilot flow rate. This behavior has been observed in piloted CH_4/air flames [34, (see p. 164)]. The Favre mean dissipation profiles show qualitatively similar trends, with peak values increasing for the first three flames in each series and then decreasing. This trend is the most obvious for the $\phi_j = 2.1$ series (Fig. 10). Note that the three lowest pilot flow rates for this series are lower than for the $\phi_j = 3.2$ and $\phi_j = 2.5$ flame series, and the last flame in Fig. 10 displays the highest probability of local extinction of all flames studied in the present work.

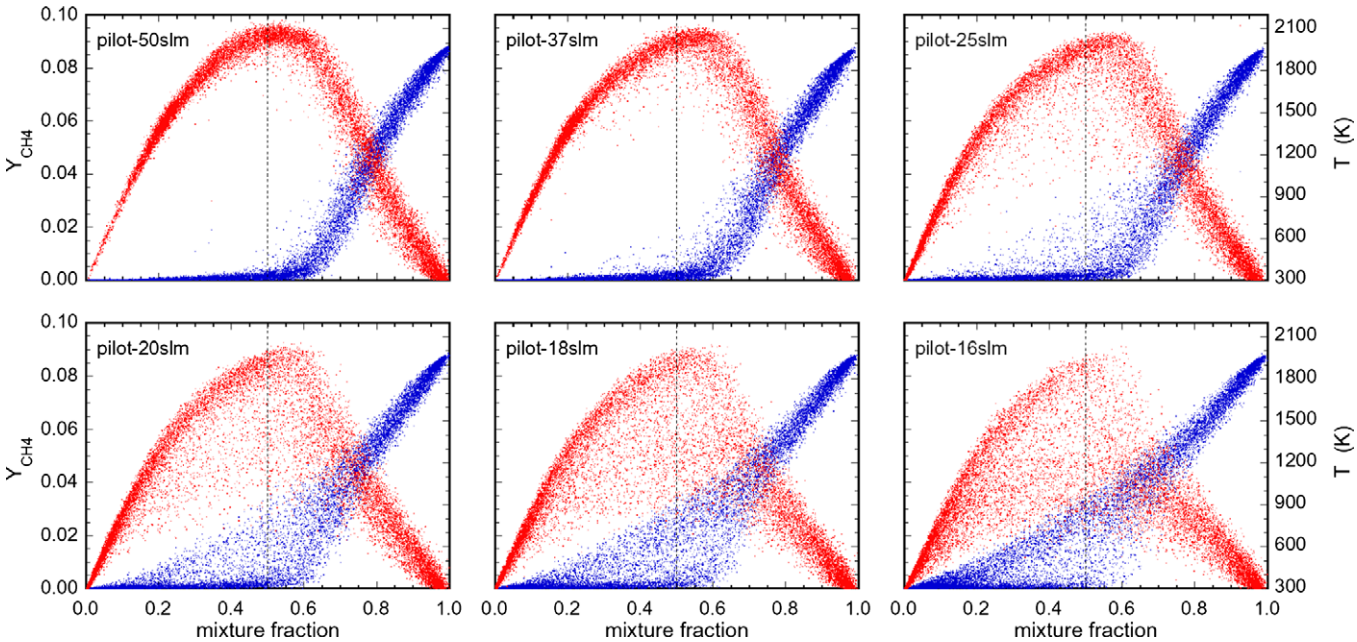

Fig. 9. Scatter plots of CH_4 mass fraction and temperature at $x/d = 4$ in each of the six flame with $\phi_j = 2.1$. The stoichiometric mixture fraction $\xi_{st} = 0.50$ is marked by a dashed line in each graph.

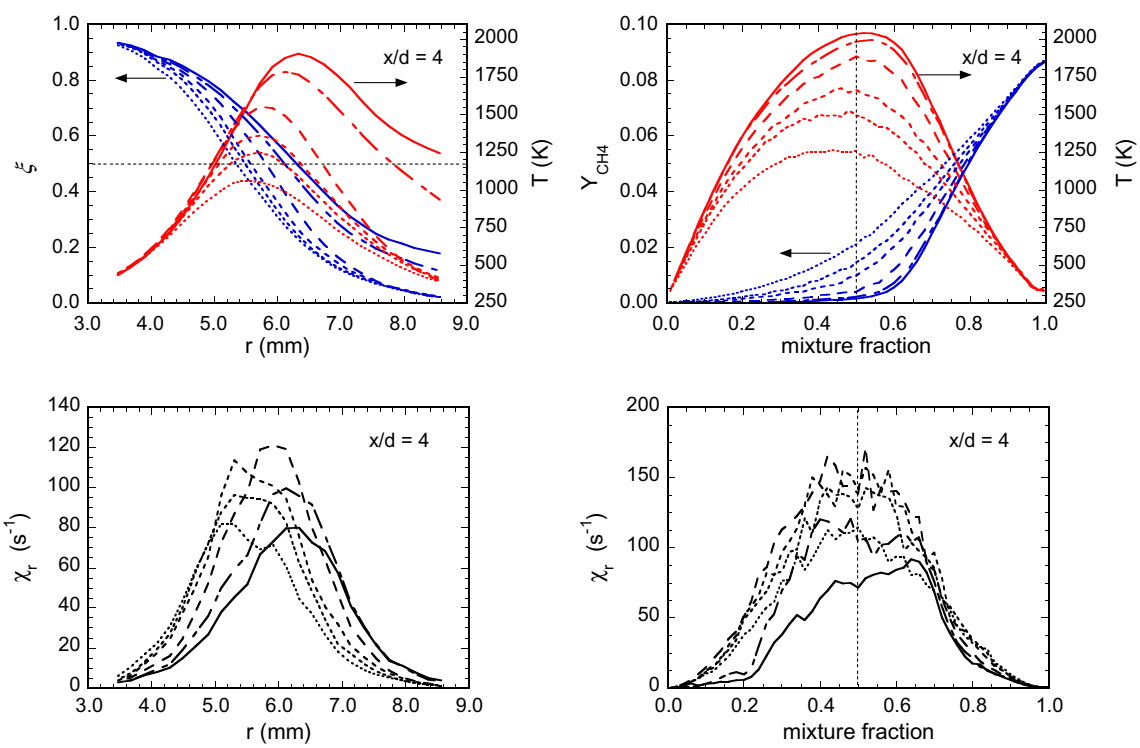

Fig. 10. Favre average radial profiles (left column) and density-weighted conditional means (right column) including mixture fraction, temperature, CH_4 mass fraction, and resolved scalar dissipation at $x/d = 4$ in the $\phi_j = 2.1$ flame series. Line types correspond to the six pilot flow rates listed in Table 1: 50 lpm (solid) down to 16 lpm (dotted). The stoichiometric mixture fraction $\xi_{st} = 0.50$ is marked by dashed lines.

A convenient way to quantify and compare effects of finite rate chemistry and local extinction in nonpremixed and partially premixed flames is to define a burning index, *BI*, based on selected scalars, (e.g., [14,19]). For example, the local instantaneous burning index based on temperature would be

$$BI_T = (T - T_0)/(T_{\mathrm{ref}|\xi} - T_0), \tag{2}$$

where T_0 is room temperature in the present experiments and $T_{\mathrm{ref}|\xi}$ is a reference temperature representing a fully burnt state at a given mixture fraction. The reference is sometimes taken to be the adiabatic equilibrium state. However, for hydrocarbon flames it is typically taken from a laminar flame calculation at low strain rate because equilibrium is not a useful reference for fuel-rich conditions.

The selection of a specific laminar flame reference for the present jet flames is somewhat arbitrary for two reasons. First, partially premixed methane flames at low strain rates have a double reaction zone, with the main reaction zone at the stoichiometric mixture fraction and a fuel-rich 'premixed' reaction zone that is supported by the presence of the main reaction zone. The structure of the fuel-rich reaction zone and its separation distance from the stoichiometric reaction zone is strongly influenced by both strain rate and radiation [35,36]. Second, it is not clear how important the effects of differential diffusion are in the present jet flames. It is common to assume unity Lewis number (equal diffusivities for species and thermal energy) in the modeling of turbulent flames. Results from studies involving multiscalar measurements in downstream portions of fully connected turbulent jet flames have been well represented by laminar flame calculations with all species diffusivities set equal to the thermal diffusivity (e.g., [7,37]). However, previous studies of nonpremixed or partially premixed jet flames have also shown that differential diffusion effects can be important in the near field of jet flames, particularly at low Reynolds number [7]. The present flames have reasonably high Re, but measurements are from the near field, so differential diffusion can not be discounted.

Effects of strain rate and differential diffusion on laminar flame structure are illustrated in Fig. 11, which shows results for major species mass fractions and temperature from opposed flow calculations at three strain rates (25, 100, and 400 s^{-1}) and using multi-component transport (Fig. 11a) or equal diffusivities (Fig. 11b). Fuel-side composition corresponds to the $\phi = 2.5$ jet flame. Both 25 s^{-1} calculations (solid curves) show distinct double reaction zones. The choice of molecular transport model affects all parts of the flame, but differences are most obvious at very rich conditions where the curves of species mass fractions vs. mixture fraction show significant bending due to preferential diffusion of H_2 when molecular transport is calculated in the normal way.

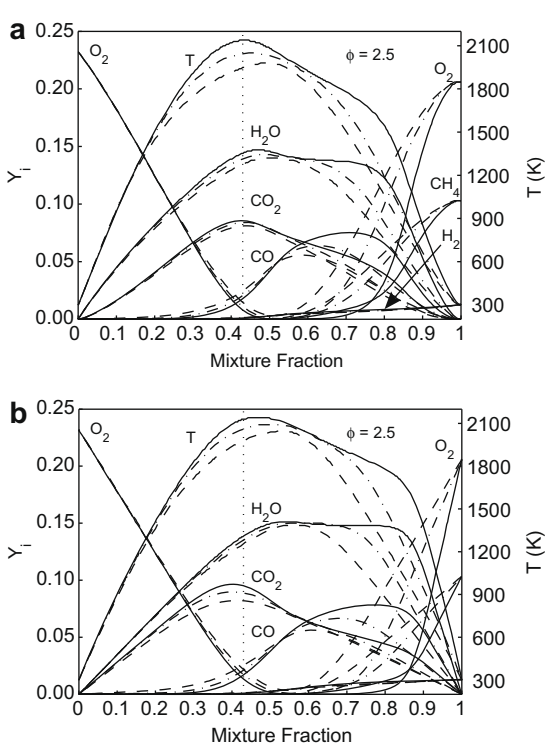

Fig. 11. Calculated mass fractions and temperature vs. mixture fraction for partially premixed laminar flames with fuel side equivalence ratio $\phi = 2.5$ at strain rates of 25, 100, and 400 s^{-1} using (a) normal molecular transport model and (b) equal thermal and mass diffusivity.

Measured conditional means of T, Y_{H_2O}, and Y_{CO_2} are shown in Fig. 12 for the three fully connected jet flames (50 lpm pilot flows) and compared with laminar flame curves for strain rates of 100 and 400 s^{-1} using the two different transport assumptions. The turbulent flame measurements show no indication of a fuel-rich reaction zone. Apparently, the high mixing rates (high scalar dissipation rates) in the fuel-rich portion of the shear layer prevent the relatively slow reactions from progressing to any significant degree at these conditions. However, there is some bending of the curves of conditional mean species mass fraction vs. mixture fraction at fuel-rich conditions, which suggests an influence of preferential H_2 diffusion. At stoichiometric and fuel-lean conditions the measured conditional means tend to fall between the laminar curves for the two transport assumptions. This suggests that turbulent transport is not dominate, allowing preferential molecular diffusion to have some influence on the overall flame structure. It should be said, however, that uncertainty in mixture fraction and mass fractions makes it difficult to draw firm quantitative conclusions on the relative importance of molecular and turbulent transport in these flames.

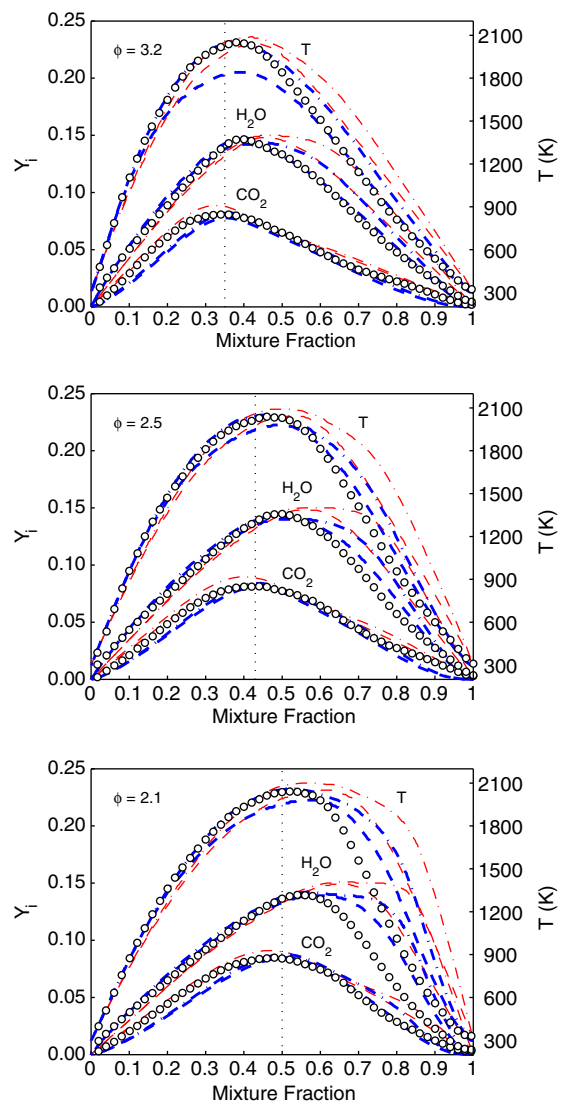

Fig. 12. Density-weighted conditional means of T, Y_{H_2O}, and Y_{CO_2} from the three fully connected turbulent flames (highest pilot flows) compared to corresponding opposed flow flame calculations including differential molecular diffusion (blue) or equal diffusivities (red) at strain rates of 100 s^{-1} (dot-dash) and 400 s^{-1} (dash). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 13 compares measured and calculated results for the differential diffusion parameter, $z = \xi_H - \xi_C$, the difference between elemental mixture fractions for hydrogen and carbon. Results for $\phi = 2.5$ are qualitatively similar to those from the other two flames. If transport where completely dominated by turbulent advection, then measured z would be zero for all values of mixture fraction. The turbulent flame measurements generally show z values of smaller magnitude than in the laminar flame calculations. Conditional mean z is slightly negative for very rich conditions, indicating preferential diffusion of H_2 away from the jet fluid at the rich edge of the mixing layer. Measured z values near stoichiometric are similar to those in the laminar flames. However, the variations in z across mixture fraction space are smaller than in the calculations. As the probability of local extinction increases within the turbulent flame series, the measured z values tend to decrease in magnitude. There are a few possible physical effects and experimental artifacts that might contribute to the observed trends in the measurements. First, as the pilot flow rate decreases, mixing rates increase across stoichiometric and lean samples (Fig. 8), and this is generally expected to reduce the relative importance of differential diffusion. Second, with increasing probability of extinction the fraction of unburnt CH_4 increases, and part of the differential diffusion effect comes from the separation and subsequent diffusion of hydrogen away from carbon in the reaction zone. Third, the level of flame luminosity decreases as the probability of local extinction increases, and this could cause a systematic shift in conditional mean species mass fractions due to errors in corrections for luminosity.

Based on the observations from Figs. 12 and 13 that differential diffusion has some influence on scalar structure in these jet flame, the relatively low strain laminar flame ($25\ s^{-1}$) with differential transport was selected as a reference for calculation of the burning index for temperature. Conditional mean results for BI_T from all three flame series (18 flames in total) are plotted in Fig. 14. The fully connected flames at each value of the jet equivalence ratio each show BI_T near unity from lean conditions across to mixture fraction values roughly 1.3 times stoichiometric. The modest dip in burning index centered near 0.8 ξ_{st} may well be due to systematic error of a few percent in mixture fraction within that range of conditions. BI_T is quite low in fuel-rich samples, even in the fully burning jet flames, because mixing rates are still sufficiently high to prevent formation of the second reaction zone that occurs in the low strain rate laminar flame calculations. The effect is most severe in the $\phi = 2.1$ series, where BI_T drops below 0.2 even for the fully connected flame with the highest pilot flow rate. As pilot flow rate is decreased in each flame series BI_T shifts downward across a broad interval in mixture fraction. The relative changes in BI_T between flames in each series should be insensitive to uncertainties in mixture fraction or individual species. The shift from flame to flame should also be insensitive to uncertainty in the measured indicators of differential diffusion. Therefore, prediction of this progression should be a good test for turbulent combustion models that include strong effects of finite rate chemistry.

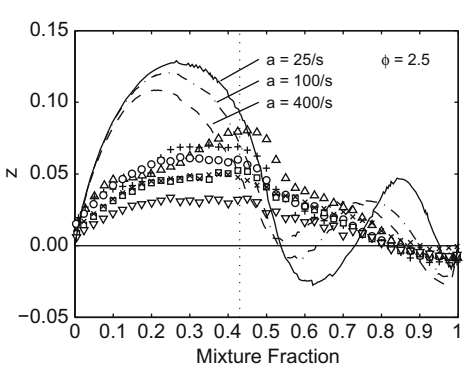

Fig. 13. Density-weighted conditional means of the differential diffusion parameter, z, from the six flames in the series with $\phi = 2.5$ are compared to results from opposed flow flame calculations at the labeled strain rates. Symbols in the order △ + ○ □ × ▽ correspond to decreasing pilot flow, as listed in Table 1, corresponding to increasing probability of local extinction.

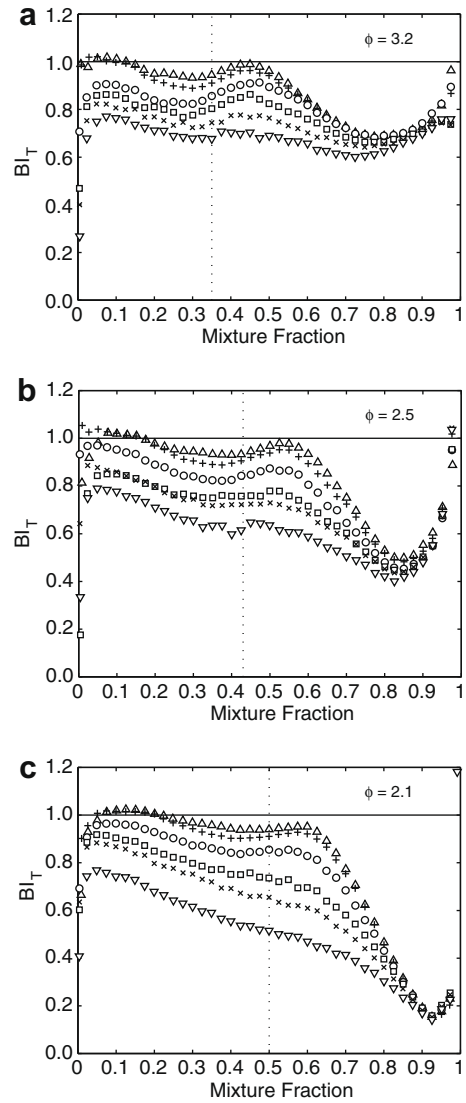

Fig. 14. Density-weighted conditional means of the burning index for temperature, BI_T, in the three series of flames. All results are from $x/d = 4$. Symbols in the order △ + ○ □ × ▽ correspond to decreasing pilot flow rates, as listed in Table 1, corresponding to increasing probability of local extinction and lower values of BI_T.

4. Conclusions

Measurements of temperature and mass fractions of major species were obtained in piloted partially premixed jet flames of CH_4/H_2/air, using the combination of line-imaged Raman/Rayleigh scattering and line-imaged CO-LIF. Objectives were to extend the basis for model validation by investigating local extinction in piloted jet flames with higher Reynolds number (60,000 and 67,000) than previously studied and by including multiple steps in the progression of local extinction. Three levels of partial premixing were considered, with stoichiometric values of mixture fraction $\xi_{st} = 0.35$, 0.43, and 0.5. For each mixture a series of six flames was studied,

with the central jet velocity held constant while the flow rate to the annular pilot flame was progressively reduced. Each series began with a fully connected flame and stepped through increasing levels of local extinction. Local extinction in these flames occurred relatively close to the jet exit in a region centered roughly four nozzle diameters downstream of the jet exit ($x/d = 4$). Measurements were obtained at $x/d = 1$ and $x/d = 2$ in order to characterize the pilot entrainment region and at $x/d = 4$ to investigate the region of local extinction.

These CH_4/H_2/air flames produced lower levels of hydrocarbon fluorescence than previously studied CH_4/air flames. However, chemiluminescence was more intense due to high rates of mixing and reaction. Corrections for this luminous background proved difficult, leading to increased uncertainty in the Raman measurements. In spite of this limitation, overall results on the progression of local extinction are considered robust, particularly when expressed as relative changes in the burning index based on temperature.

In each flame series results show that reduction in pilot flow rate allows a more rapid entrainment of hot products from the pilot flame into the mixing layer at the edge of the high velocity jet. Once the pilot products have been entrained, the scalar dissipation rate within the reaction zone increases to a critical level, causing local extinction. Dissipation measurements are not fully resolved but still provide qualitative insights on turbulence–chemistry interaction in the present flames. With increasing probability of local extinction the resolved density-weighted conditional mean scalar dissipation levels off and then actually decreases. This may be due to the influence of decreased temperature on diffusivity.

Conditional mean scalar measurements were compared with results from laminar flame calculations. These highly strained turbulent flames showed no indication of a double reaction zone, as appears in partially premixed laminar flames at low strain rates. However, in spite of the relatively high Reynolds numbers of the turbulent flames, results indicate that differential diffusion has some influence on scalar structure at the near-field locations studied in the present work. This is a disadvantage of having the extinction region close to the jet exit.

The absence of significant reaction progress at fuel-rich conditions was apparent in results for the conditional mean burning index based on temperature, BI_T. The results showed a progressive decrease in BI_T as pilot flow rate was decreased, and the effect was broadly spread across the full mixture fraction range. This measure of the degree of local extinction should allow for convenient comparison with model calculations of the present jet flames, and prediction of the progression in the burning index should be a good test for turbulent combustion models that include strong effects of finite rate chemistry.

Acknowledgments

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